

⑤ $L = \{ a^i b^j c^k d^m \mid i \geq m \text{ y } j > k \}$

Paso 1: Símbolos inútiles
No hay símbolos inútiles

$$S \rightarrow aSD \mid AP$$

$$P \rightarrow bPC \mid B$$

$$A \rightarrow Aa \mid a$$

$$B \rightarrow Bb \mid b$$

$$D \rightarrow d \mid \lambda$$

$$C \rightarrow c \mid \lambda$$

Paso 2: Eliminar proyecciones con λ

$$S \rightarrow aS \mid aSd \mid AP$$

$$P \rightarrow bP \mid bPc \mid B$$

$$A \rightarrow Aa \mid a$$

$$B \rightarrow Bb \mid b$$

Paso 3: Eliminar proyecciones unitarias
(P, B)

$$S \rightarrow aSd \mid AP$$

$$P \rightarrow bP \mid bPc \mid Bb \mid b$$

$$A \rightarrow Aa \mid a$$

$$B \rightarrow Bb \mid b$$

⑥ $L = \{ a^i (b+c)^k a^j \mid k \neq i + j \text{ y } i, j, k \geq 0 \}$

$$S \rightarrow APQ \mid PQA \mid APQA \mid PBO$$

$$P \rightarrow aPZ \mid \lambda$$

$$Q \rightarrow ZYa \mid \lambda$$

$$Z \rightarrow b \mid c$$

$$B \rightarrow bB \mid b \mid cB \mid c$$

$$A \rightarrow Aa \mid a$$

Paso 1: Eliminar símbolos inútiles

No hay $\forall S \in G$ S es Generador y Alcanzable

Paso 2: Eliminar proyecciones con λ

$$S \rightarrow A \mid AP \mid APQ \mid AQ \mid PA \mid QA \mid PQA \mid APQA \mid PB \mid BQ \mid PBQ \mid B \mid APA \mid AQA$$

$$P \rightarrow aZ \mid aPZ$$

$$Q \rightarrow ZYa \mid Za$$

$$Z \rightarrow b \mid c$$

$$B \rightarrow bB \mid b \mid cB \mid c$$

Paso 3: Eliminar proyecciones unitarias

$$S \rightarrow Aa \mid a \mid AP \mid APQ \mid AQ \mid PA \mid QA \mid PQA \mid APQA \mid PB \mid BQ \mid PBQ \mid bB \mid b \mid cB \mid c \mid APA \mid AQA$$

$$P \rightarrow aZ \mid aPZ$$

$$Q \rightarrow ZYa \mid Za$$

$$Z \rightarrow b \mid c$$

$$B \rightarrow bB \mid b \mid cB \mid c$$

⑦ $L = \{ a^i b^j c^k a^m b^p \mid i \geq j+k \text{ y } m=p \}$

$$S \rightarrow PQ$$

$$P \rightarrow aPc \mid R$$

$$R \rightarrow aRb \mid \lambda$$

$$Q \rightarrow aQb \mid \lambda$$

Paso 1: Símbolos inútiles

$\forall S \in G$

$S \in \{\text{Generadores}\}$

y

$S \in \{\text{Alcanzables}\}$

Paso 2: Eliminar proy con λ

$$S \rightarrow P \mid Q \mid PQ \mid \lambda$$

$$P \rightarrow ac \mid aPc \mid R$$

$$R \rightarrow ab \mid aRb$$

$$Q \rightarrow ab \mid aQb$$

Paso 3: Elimine proyecciones unitarias

$$S \rightarrow ac \mid aPc \mid ab \mid aRb \mid aQb \mid \lambda \mid PQ$$

$$P \rightarrow ac \mid aPc \mid ab \mid aRb$$

$$R \rightarrow ab \mid aRb$$

$$Q \rightarrow ab \mid aQb$$